INCH-POUND
MIL-DTL-24332A(SH)
2 April 2019
SUPERSEDING
MIL-V-24332(SHIPS)
1 July 1968

DETAIL SPECIFICATION

VALVES, ANGLE RELIEF

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained, single-seated, and spring-loaded angle relief valves (see 6.3.3) of totally-enclosed, spring-pressure tight construction for liquid service (except hydraulic oil), where the inlet pressure is sensed under and directly operates the spring-loaded disc (see 6.3.7).

1.2 Classification. Relief valves (see 6.3.10) are of the following grades, intended use, and system as specified (see 6.2):

1.2.1 Grades. Grade refers to the material of the body, bonnet, and yoke.

Grade	PIN Code	Description
A	A	Ferrous
B	B	Nonferrous

1.2.2 Intended service.

Intended Service	PIN Code
Flammable or combustible fluid	1
Non-flammable or non-combustible fluid	2
Seawater service	3

1.2.3 Systems.

System	PIN Code
Closed system	0
Not closed system	1

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). The PINs for check valves as created as follows:

M	24332	-	X	X	X
Prefix for Military Specification	Specification Number	-	Grade (see 1.2.1)	Intended Service (see 1.2.2)	System (see 1.2.3)

Example: M24332-A-1-1 is a grade A body, bonnet, yoke intended for flammable or combustible fluid.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought UNS N05500

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Service

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing General Specification for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-DTL-24227 - Fittings and Flanges, Cast Bronze, Silver-Brazing Suitable for Ultrasonic Inspection

MIL-PRF-32187 - Gasket, Hybrid, Flat and Raised Face Flange, Fire Resistant, Shipboard Use

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-798 - Design Evaluation Requirements for Valves for Naval Shipboard Use

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
S9086-RK-STM-010/505	-	NSTM Chapter 505, Piping Systems
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process
T9074-AS-GIB-010/271	-	Requirements of Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC	-	ASME Boiler and Pressure Vessel Code, Section VIII, Rules for Construction of Pressure Vessels
ASME B16.5	-	Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24 Metric/Inch Standard

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A105/A105M	-	Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A193/A193M	-	Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications
ASTM A194/A194M	-	Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both
ASTM A216/A216M	-	Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
ASTM A217/A217M	-	Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service
ASTM A227/A227M	-	Standard Specification for Steel Wire, Cold-Drawn for Mechanical Springs
ASTM A229/A229M	-	Standard Specification for Steel Wire, Quenched and Tempered for Mechanical Springs
ASTM A230/A230M	-	Standard Specification for Steel Wire, Oil-Tempered Carbon Valve Spring Quality
ASTM A231/A231M	-	Standard Specification for Chromium-Vanadium Alloy Steel Spring Wire
ASTM A232/A232M	-	Standard Specification for Chromium-Vanadium Alloy Steel Valve Spring Quality Wire
ASTM A276/A276M	-	Standard Specification for Stainless Steel Bars and Shapes

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ASTM A313/A313M	-	Standard Specification for Stainless Steel Spring Wire
ASTM A351/A351M	-	Standard Specification for Castings, Austenitic, for Pressure-Containing Parts
ASTM A479/A479M	-	Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels
ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM A582/A582M	-	Standard Specification for Free-Machining Stainless Steel Bars
ASTM A689	-	Standard Specification for Carbon and Alloy Steel Bars for Springs
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B164	-	Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B166	-	Standard Specification for Nickel-Chromium-Iron Alloys (UNS N06600, N06601, N06603, N06690, N06693, N06025, N06045, and N06696), Nickle-Chromium-Cobalt-Molybdenum Alloy (UNS N06617), and Nickel-Iron-Chromium-Tungsten Alloy (UNS N06674) Rod, Bar, and Wire
ASTM B369	-	Standard Specification for Copper-Nickel Alloy Castings
ASTM B637	-	Standard Specification for Precipitation-Hardening and Cold Worked Nickel Alloy Bars, Forgings, and Forging Stock for Moderate or High Temperature Service
ASTM F467	-	Standard Specification for Nonferrous Nuts for General Use
ASTM F468	-	Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS5699	-	Nickel Alloy, Corrosion and Heat Resistant, Wire, 72Ni – 15.5Cr – 0.95Cb – 2.5Ti – 0.70Al – 7.0Fe, Spring Temper, Precipitation Hardenable
SAE J1926/1	-	Connections for General Use and Fluid Power - Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing - Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials of construction. Materials shall be as specified in table I. All materials shall be suitable for the intended pressure and shall be selected to prevent galling, seizing, or excessive wear between parts. Clearances shall be such as to prevent interferences due to thermal expansion. Welding, weld repair, and hardfacing shall be in accordance with S9074-AR-GIB-010/278. This specification is not intended to be restrictive, providing proposed alternate materials will provide satisfactory service. The use of any alternate material shall require Naval Surface Warfare Center (NSWC) Philadelphia (Code 414) approval.

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3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Prohibited materials. The valves shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

TABLE I. List of materials.

Name of part	Grade A Applicable documents	Grade B Applicable documents
Body, bonnet, and yoke	ASTM A105/A105M, grade II ASTM A216/A216M, grade WCB ASTM A217/A217M, grade WC1 and WC6 ASTM A351/A351M, grade CF3, CF3M, CF8, and CF8M	ASTM B61 ASTM A494/A494M, grade M30H ASTM B369
Disc and seat ring	ASTM A276/A276M, type 302, 304, 316, 431, and 440 ASTM A582/A582M, type 303 ASTM A351/A351M, grade CF3, CF3M, CF8, and CF8M ASTM B164 ASTM A494/A494M	
Stem	ASTM A276/A276M or ASTM A479/A479M, type 304 and 316 ASTM B164 ASTM A582/A582M, type 303 and 303Se	ASTM B164 ASTM A276/A276M or ASTM A479/A479M, type 304 and 316 ASTM A582/A582M, type 303 and 303Se QQ-N-281
Springs	ASTM A231/A231M ASTM A232/A232M ASTM A230/A230M ASTM A313/A313M ASTM A689 QQ-N-286 ASTM A227/A227M ASTM A229/A229M	ASTM B164 ASTM B166 ASTM A313/A313M ASTM A276/A276M QQ-N-286 SAE AMS5699 ASTM B637
Body bolts and nuts	ASTM A193/A193M, grade B7 ASTM A194/A194M, grade 2H	QQ-N-286, class A ASTM F467, Alloy 400/405 ASTM F468, Alloy 400 MIL-DTL-1222, grade 500
NOTE: 1/ See 3.3.6 for material limitations based on application.		

3.3 Design and construction.

3.3.1 Pressure-temperature ratings. As specified (see 6.2), the design and pressure-temperature rating for valves of this specification shall be in accordance with ASME B16.5.

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3.3.2 Inlet ratings and end connections. The inlet ratings for grade A valves shall be in accordance with ASME B16.5. Grade A valves shall have flanged inlet and outlet connections in accordance with ASME B16.5. Unless otherwise specified (see 6.2), the inlet ratings and end connections for grade B valves shall be in accordance with table II. When union end connections are specified (see 6.2), the union nuts and tailpieces shall be furnished. Only the pertinent dimensions of the documents listed in table II apply.

TABLE II. Inlet ratings and end connections for grade B valves.

Nominal pressure rating (psig)	Union end 1/	Flanged end
150	MIL-DTL-1183	MIL-PRF-20042
250	MIL-DTL-1183	MIL-PRF-20042
400	MIL-DTL-1183	MIL-PRF-20042

NOTE:
1/ On applications where ultrasonic test inspection of sil-brazed joint is required, MIL-F-24227 shall be used in lieu of MIL-DTL-1183.

3.3.3 Outlet ratings. Unless otherwise specified (see 6.2), the outlet portion shall be designed to withstand 150 percent of the maximum system backpressure or 100 pounds per square inch gauge (psig), whichever is higher.

3.3.4 Body construction. The valve shall be designed and constructed so that the seat (see 6.3.11) shall not become distorted relative to the disc, and valve operation is not adversely affected by internal pressure.

3.3.5 Bonnet construction. The bonnet (see 6.3.6) or spring housing shall be attached to the body with bolted flanges or threaded union connection. Bearing surfaces of nuts and bolts, and their respecting mating surfaces on the valve, shall be machine finished.

3.3.6 Internal trim. Unless otherwise specified (see 6.2), valve seats may be separate or integral with the valve body. Valves without separate seat rings shall have sufficient metal, at the seat section, to permit installation of the separate seat ring if required as a service repair. Seat rings shall be threaded in and shall be easily replaceable, using hand tools, after extended service. The seat ring shall shoulder against the body to provide a positive pressure tight joint in which the threads are not used to seal. Where a nonmetallic seal is used between the body and seat ring, a gland or cavity shall be provided in either the body or seat ring to ensure proper retention of the seal. The stem or stem and disc assembly shall be top guided (all guides shall be on the discharge side of the seat). Guiding surfaces (bushings and posts) shall have the proper hardness, finish, concentricity, parallelism, clearances, length, and rigidity to prevent binding or seizing and to ensure proper seating under all operating conditions. These alignment requirements shall be maintained with interchangeable parts and under any tolerance stack-up condition.

3.3.6.1 Flammable or combustible fluid. Valves intended for flammable or combustible liquid service (see 6.2) as defined by S9086-RK-STM-010/505 shall have metal-to-metal seats. Gaskets shall be soft metal or in accordance with MIL-PRF-32187. For seals at threaded connections (SAE J1926/1), an elastomeric seal is acceptable if the fitting or washer has metal-to-metal contact on the spotface beyond the O-ring seal.

3.3.6.2 Non-flammable or non-combustible fluid. Where valves are intended for services where the operating temperatures do not exceed 250 °F (see 6.2), a nonmetallic seating feature shall be incorporated in either the disc or seat.

3.3.6.3 Seawater service. Valves intended for seawater service or that have internals exposed to seawater (see 6.2) shall have nickel-copper alloy stems, seats, discs, disc nuts, hinge pins, and side plugs, as applicable. For valves in sizes 2½ inches and above, discs with a nickel-copper alloy facing of ⅛ inch minimum thickness are acceptable in lieu of solid nickel-copper alloy. Nickel-copper aluminum or nickel-copper-silicon alloy may be used as an alternative to nickel copper. Bronze valves for other than seawater service may have bronze stems, seats, discs, disc nuts, hinge pins, and side plugs, as applicable.

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3.3.7 Springs. Springs (see 6.3.15) shall be designed so that they will not be fully compressed or stressed beyond two-thirds of the yield strength of the spring material during any normal operation or adjustment of the valve. When removed and compressed solid, the springs shall not exhibit a permanent set exceeding 0.010 inch per inch of free spring length, measured 10 minutes after release of the spring. Spring ends shall be squared and ground.

3.3.8 Threads. All threads shall conform to FED-STD-H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used.

3.3.9 Interchangeability. All parts having the same manufacturer's part number should be directly interchangeable with each other with respect to installation and performance and should not require selection, fitting, or machining of any kind. Where machining is required after installation of a seat ring or guide in order to maintain critical concentricity or alignment dimensions, detailed instructions may be provided with each repair part.

3.3.10 Shipboard maintainability. Where the intended use of the valve is a shipboard system installation, master drawings shall be as specified (see 6.2 and 6.5).

3.3.11 Hand lifting device. Valves shall be designed so that they may be operated by hand for testing purposes with an inlet pressure of at least 75 percent of the set pressure (see 6.3.12). The necessary lever or tool shall be furnished as part of each valve.

3.3.12 Stuffing boxes. A stuffing box on the valve stem shall not be permitted. A stuffing box on the shaft of the hand lifting device, which will have no effect on the relief valve setting, shall be required.

3.3.13 Gagging device. Where required for system test purposes, a gagging device (see 6.3.8) shall be supplied with the valve, as specified (see 6.2). Valves shall be capable of being gagged without alteration of the set point. The gagging screw shall be designed to discourage the use of wrenches when gagging the valve. The gagging device shall be designed to minimize the possibility of overlooking its removal after test and shall include a tag or other warning to this effect.

3.3.14 Accessibility. Valves shall be accessible for adjustment and repair without requiring removal from the line, except in the case of full nozzle designs.

3.3.15 Valve adjustment. Means shall be provided for adjusting the setting with the valve under pressure. The adjusting screw shall have right hand threads so that clockwise rotation increases the set pressure. The adjusting device shall be provided with a locknut and cap, or other suitable means, to prevent accidental change of adjustment.

3.3.16 Envelope integrity. When the body and bonnet walls are specified to be solid (see 6.2), plugged drain connections or other penetrations are not permitted.

3.4 Performance. Unless otherwise specified (see 6.2), the performance requirements shall be as follows:

3.4.1 Set pressure range. For set pressures ranges (see 6.3.13) up to 250 psig, the set pressure shall be adjustable over a range of ±10 percent without requiring replacement of any internal parts. For set pressures over 250 psig, the set pressure shall be adjustable over a range of ±5 percent without requiring replacement of any internal parts.

3.4.2 Operation. Where properly installed in accordance with the limitations specified by the valve manufacturer, relief valves shall operate over the entire flow range without chatter. Valve opening and closing may be gradual and in proportion to the inlet pressure.

3.4.3 Set pressure tolerance. The set pressure tolerance (see 6.3.14), shall not exceed ±2 pounds per square inch (psi) or 2 percent of the set pressure, whichever is greater.

3.4.4 Accumulation. When specified (see 6.2) valves shall be sized to pass the capacity specified without permitting the inlet pressure to exceed the maximum allowable accumulation (see 6.3.1) pressure specified on figure 1.

3.4.5 Blowdown. Valves shall operate satisfactorily with a blowdown (see 6.3.4) not less than the minimum allowable blowdown pressure (see 6.3.5) permitted by figure 1.

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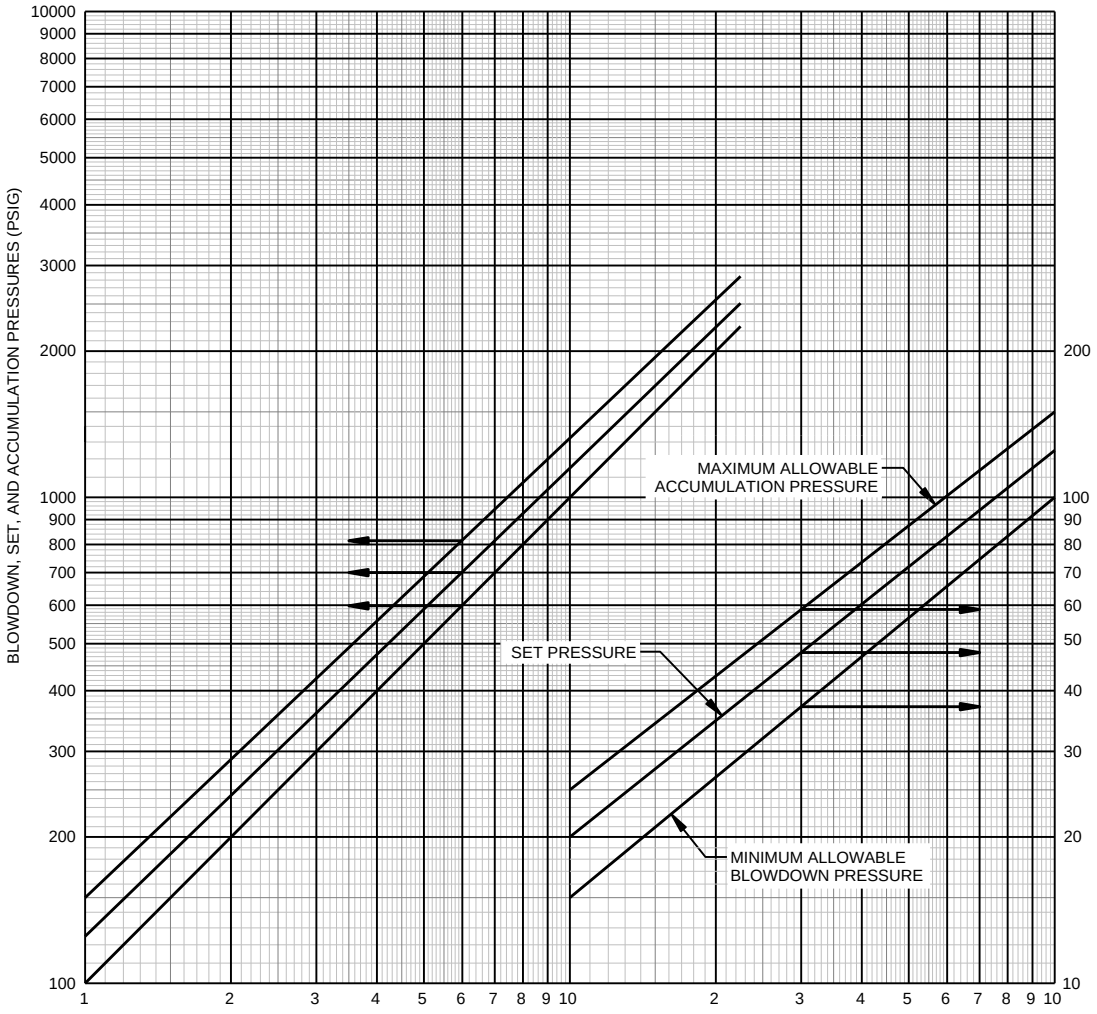


FIGURE 1. Relief valve characteristics.

3.4.6 Seat tightness. Valves with a nonmetallic seating feature shall seat tightly (i.e., shall have no visible leakage over a 3-minute period) with an inlet pressure equal to the minimum allowable blowdown pressure permitted by figure 1 for the applicable nominal set pressure (see 6.3.9). Valves incorporating metal-to-metal seating shall not have more than 10 cubic centimeters per hour per inch of seat diameter leakage.

3.4.7 Installation limitations. Valve operation shall not be adversely affected by an inlet pressure loss up to 25 percent of the relief valve blowdown, a discharge pressure buildup to 10 percent of the nominal set pressure, or both. Information stating that installation will subject the valve to greater piping restrictions shall be as specified (see 6.2).

3.5 Shock and vibration. Unless otherwise specified (see 6.2) valves shall be designed to meet the grade A, class I high-impact mechanical shock requirements of MIL-DTL-901 (see 4.5) and the type I environmental vibration requirements of MIL-STD-167-1 up to and including 25 cycles per second (see 4.6).

3.6 Marking.

3.6.1 Body markings. The valve manufacturer's name or trademark and the body material composition shall be cast or forged integral with the valve body. The size and rating (inlet and outlet) shall be cast or forged integral with the valve body or stamped on the outside diameter of the flanges.

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3.6.2 Identification plates. Each valve shall have an identification plate permanently attached to an exposed position on the valve. The identification plate shall be made of a corrosion-resistant material and shall contain the following information or a space therefor:

a. Manufacturer's name or trademark.

b. Rated capacity at the applicable setting and accumulation.

c. Body material composition.

d. Service fluid.

e. Set pressure (psig), blowdown pressure (psig), and accumulation pressure (see 6.3.2).

f. Range of set pressure adjustment available with installed spring.

g. Installation limitations of valve (maximum permissible inlet pressure loss and maximum permissible backpressure buildup for which valve is designed).

h. Manufacturer's model or part number and drawing number.

i. Space for a 9-digit Allowance Parts List (APL) number.

3.7 Workmanship. Valves shall be free from defects that affect their appearance or operation. Castings shall be clean, sound, and free from hard spots, porosity, cracks, and other injurious defects. Castings shall be smooth and well-cleaned, both inside and outside, and all fins and roughness shall be removed. Visual inspection of castings shall be in accordance with 4.4.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the first valve of each lot, or when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.4.1 and the tests of 4.4, 4.5, and 4.6. First article tests may be waived provided certified test data is available showing that the same basic design and size has previously passed these tests.

4.2.1 Lot. All relief valves of the same size, design, pressure rating, setting, and capacity offered for delivery at one time shall be considered one lot.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4.1 and the tests of 4.4.2, 4.4.3, 4.4.5, and 4.4.6. Each valve of each lot shall be subjected to conformance inspection. Any discrepancies or failures shall be corrected prior to acceptance.

4.4 First unit examination and performance test outline. Unless otherwise specified (see 6.2), the performance requirements verified in 4.4.3 are based on figure 1. Figure 1 plots the nominal set point and accumulation pressures based on the reseat pressure. The two limiting parameters are the maximum accumulation and blowdown curves. The valve's actual set point can vary per the set pressure range as long as the blowdown and accumulation are within that envelope (see 3.4.7).

4.4.1 Visual examination. Each valve, without disassembly, shall be visibly examined for compliance with approved design drawings and the requirements specified in 3.2, 3.3, 3.6, 3.7, and 4.7. Any redesign or modification of the contractor's standard product to comply with specified requirements shall receive particular attention for adequacy and suitability. Visual inspection and acceptance criteria of castings shall be in accordance with S9074-AR-GIB-010/278. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.2 Performance proof test. A proof test shall be completed to test strength and soundness of pressure containing envelope.

4.4.2.1 Test conditions. The following test conditions shall be performed:

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a. Valve gagged shut or disc and spring assembly removed, and seat blanked off.

b. Water at 1.5 times rated pressure to be applied to inlet and maintained for 3 minutes.

c. Water at pressure defined by 3.3.3 applied to outlet.

4.4.2.2 Acceptance criteria. No external leakage, permanent deformation, or structural failure shall be present.

4.4.3 Performance set pressure, blowdown, and seat tightness test. Set pressure, blowdown, and seat tightness tests shall be performed of valve to determine set point and blowdown setting, and to test for seat tightness at the minimum allowable blowdown pressure permitted by figure 1.

4.4.3.1 Test conditions. The following test conditions shall be performed.

a. Inlet pressure increased until valve lifts.

b. Inlet pressure reduced until valve reseats.

c. Check for leakage over a 3-minute period with an inlet pressure equal to the minimum allowable blowdown pressure permitted by figure 1.

4.4.3.2 Acceptance criteria.

a. Blowdown (see 3.4.5).

b. Seat tightness (see 3.4.6).

c. No damage to seating surfaces shall be present.

d. No instability shall be accepted.

4.4.4 Performance accumulation test. To verify relief valve capacity, instability shall be detected over operational range of valve, and the extent of leakage past valve stems shall be identified.

4.4.4.1 Test condition. Inlet pressure shall be increased until valve passes rated flow.

4.4.4.2 Acceptance criteria. The valve shall pass rated flow without exceeding accumulation permitted by figure 1. No instability shall be evident in the valve. Unless otherwise specified (see 6.2), performance requirements are based on figure 1.

4.4.5 Material, chemical, and physical test. Material, chemical, and physical tests shall be performed to verify conformance of materials used to their applicable material specifications.

4.4.5.1 Test condition. As specified in the applicable material specifications, all materials used in the manufacture of valves shall be examined and material certifications verified for conformance to 3.2.

4.4.5.2 Acceptance criteria. Valves shall only be accepted when specified material and material certifications are verified for conformance to 3.2.

4.4.6 Non-destructive tests. Pressure-containing components (bodies, bonnets, covers, and discs) shall have non-destructive testing accomplished in accordance with MIL-STD-798 with valve primary service pressure rating identified in MIL-STD-798 defined as the rating or pressure class as delineated in 3.3.2.

a. Radiographic examination shall be in accordance with T9074-AS-GIB-010/271. Acceptance criteria for radiographic testing shall be as defined in S9074-AR-GIB-010/278.

b. Magnetic particle inspection shall be in accordance with T9074-AS-GIB-010/271. Acceptance criteria shall be as defined in MIL-STD-2035.

c. Dye penetrant inspection shall be in accordance with T9074-AS-GIB-010/271. Acceptance criteria shall be as defined in MIL-STD-2035.

4.4.7 Performance set pressure repeatability and endurance tests. Set pressure repeatability and endurance tests shall be performed to verify that set pressure repeatability is within allowable limits and to verify valve ability to withstand repetitive cycling.

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4.4.7.1 Test conditions.

a. Cycle valve 50 times.

b. After every 10 cycles, check for leakage.

4.4.7.2 Acceptance criteria. Any damage caused by foreign particle entrapment on the seating surfaces shall be repaired prior to acceptance of the valve.

a. Set pressure repeatability (see 3.4.3).

b. Seat tightness (see 3.4.6).

c. No damage to seating surface caused by cycling impact shall be present.

d. No instability shall be present.

4.4.7.3 Closed systems. Where relief valve discharge is installed in a closed system (see 6.2) and the discharge line is potentially filled with liquid, the repeatability and endurance test stand, as specified in MIL-STD-167-1, shall ensure that the discharge line simulates a liquid-filled condition during the test.

4.5 Shock test. Each type of valve design shall be shock tested to ensure compliance with the requirements specified in 3.5 and shall be subject to review for similarity of design and materials as the basis of approval by extension in accordance with MIL-DTL-901. Unless otherwise specified (see 6.2), successful tests will be accepted as meeting this test. Extension guidance is provided in MIL-STD-798. Planning shall include identifying range of valve family, range of valve family operating conditions, and acceptance and exit criteria. Sizes to be shock tested within a family of valves shall be in accordance with MIL-STD-798.

4.5.1 Shock test procedure. Valves shall be tested at the setting required by the application for which they are being used and pressurized on the inlet side with corresponding system operating pressure. Shock approval for subsequent application involving a different set pressure shall not require retest for shock qualification or extension action because of the different set pressure (this applies to multiple spring as well as single spring designs). A different setting and inlet pressure will not, in itself, be cause for retest for shock qualification. However, in instances where a relief valve is used in a new application or set pressure, a shock test may be required. Momentary malfunction at the time of impact blow is permitted and acceptable. Permanent deformation, misalignment, or functional impairments shall be cause for rejection.

4.5.2 Shock test examination. Each valve tested shall receive a complete visual and dimensional examination before and after shock tests. Pre-shock visual and dimensional examination shall be performed on the external surfaces only, without valve disassembly. Visual and dimensional examinations of internal and external valve surfaces and components shall be conducted upon shock test conclusion. No relaxation or deviations in post shock testing shall be permitted unless approved by NAVSEA.

4.5.3 Post shock test. The following minimum post shock tests are required in all cases. The tests shall be conducted in the order listed.

a. An external visual and dimensional check. The valve shall not be disassembled. This may include a dye penetrant or magnetic particle inspection as specified in 4.4.6 depending upon the requirements of the contract.

b. A proof test, as specified in 4.4.2. Pressure shall be maintained for at least 10 minutes.

c. A seat tightness test, as specified in 4.4.3. Pressure shall be maintained for at least 3 minutes.

d. A final visual and dimensional check, including valve internals.

4.5.4 Post shock test acceptance criteria. The valve shall pass all post shock tests. In the case of the visual and dimensional checks, any permanent damage that is detrimental to the valve shall be considered cause for rejection. This shall include any permanent deformation of any pressure containing part.

4.5.5 Adjustments. Adjustments to correct minor malfunctions during shock tests will be permitted within the scope of immediate adjustment by ship personnel (e.g., tightening a packing gland is permissible and replacing a yielded or sheared bolt is not).

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4.6 Vibration test. Each valve design shall be vibration tested in accordance with MIL-STD-167-1 under normal operating conditions to duplicate shipboard installation and shall comply with requirements specified in 3.5. Evidence of previously conducted successful test will be accepted as meeting this test (see 6.2). Extension guidance is provided in MIL-STD-167-1.

4.7 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.2 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Pressure relief valves covered by this specification are intended for overpressure protection onboard naval ships. Previously approved pressure relief valves manufactured in accordance with MIL-V-24332, dated 1 July 1968, and Amendment 2, dated 6 February 1974, are acceptable for use and may continue to be ordered as replacements for existing shipboard applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade and rating required (see 1.2).

c. When first article inspection is required (see 3.1 and 4.2).

d. Requirement for prohibited materials verification (see 3.2.2 and 4.7).

e. Maximum inlet pressure and temperature (see 3.3.1).

f. Inlet ratings and end connections for grade B valves, if other than as specified (see 3.3.2).

g. Outlet ratings, if other than as specified (see 3.3.3).

h. Valve seat construction in relation to the valve body, if other than as specified (see 3.3.6)

i. Whether the valve is intended for flammable or combustible liquid service (see 3.3.6.1),

j. Non-flammable or non-combustible liquid service (see 3.3.6.2).

k. Seawater service (see 3.3.6.3).

l. Requirements for master drawings, unless the drawings have been submitted to the Navy as part of a previous order (see 3.3.10 and 6.5).

m. If a gagging device is required to be supplied with valve (see 3.3.13).

n. When the body and bonnet walls are to be solid (see 3.3.16).

o. Performance requirements (set pressure range, set pressure tolerance, accumulation, blowdown, etc.), if other than as specified (see 3.4 and 4.4).

p. Accumulation (see 3.4.4).

q. Data on additional inlet and outlet piping restrictions that will be imposed on the valve and information stating that installation will subject the valve to greater piping restrictions (see 3.4.7).

r. Whether shock and vibration tests are required (see 3.5).

s. Acceptable criteria (see 4.4.4.2).

t. Whether the valve is installed in a closed system (see 4.4.7.3).

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u Shock test (see 4.5).

v Vibration test (see 4.6).

w. Packaging requirements (see 5.1).

x. Valve orientation for packaging (see 6.6).

y. Whether the accumulation test is required (see 6.7).

6.3 Definitions. The following definitions are applicable to this specification.

6.3.1 Accumulation. The increase in static pressure, above the set pressure, required to pass rated flow. Expressed in psi or as a percent of the set pressure.

6.3.2 Accumulation pressure. The set pressure plus the accumulation. Expressed in psig.

6.3.3 Angled relief valve. A relief valve with intake and exit ports at right angles to each other.

6.3.4 Blowdown. The decrease in pressure below the set pressure, required for the valve to reseat. Expressed in psi or as a percent of the set pressure. The accumulation and blowdown establish the operating band of relief valve at a particular setting.

6.3.5 Blowdown pressure. The set pressure minus blowdown. Expressed in psig.

6.3.6 Bonnet. A component of a direct spring valve that supports the spring.

6.3.7 Disc. A moveable component of a pressure relief valve that contains primary pressure when it rests against the nozzle.

6.3.8 Gag. A device used on reclosing pressure relief devices to prevent the device from opening.

6.3.9 Nominal set pressure. The average of the maximum set pressure and the minimum set pressure over the range for which the set pressure can be adjusted with the installed spring.

6.3.10 Relief valve. A valve used to control or limit the pressure in a system or vessel which can build up for a process upset, instrument or equipment failure or fire. Rupture disc pressure relief systems are not included in this valve type.

6.3.11 Seat. The pressure-sealing surfaces of the fixed and moving pressure-containing components.

6.3.12 Set pressure. The pressure at which the valve begins to open where there is measurable lift or discharge becomes continuous. Expressed in psig.

6.3.13 Set pressure range. The range over which the set pressure can be adjusted with the installed spring.

6.3.14 Set pressure tolerance. The permissible plus or minus deviation from the specified set pressure. Expressed in psi or as a percent of set pressure.

6.3.15 Spring. The element in a pressure relief valve that provides the force to keep the disc on the nozzle.

6.4 First article waiver. When first article inspection is required (see 6.2), the contracting officer should provide guidance to offerors as to whether or not the item should be a first article sample (see 3.1, 4.4, 4.5, and4.6) as well as the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements of examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitation.

6.5 Master drawings. Detailed drawings that are sufficient to permit evaluation of the design should show the following:

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a. A sectional assembly of the valve and details of the seat, disk, and stem assembly.

b. Finishes of all guiding and seating surfaces.

c. Bill of materials listing specification, grade, condition, and any other data required to fully identify the properties of the materials proposed.

d. Installation dimensions, end connection detail, and clearance dimensions required for disassembly.

e. Performance characteristics, estimated weight, and any limitations on installation.

f. Note any previous shock, vibration, or first unit test approval.

g. Note where machining is required after seat ring or guide installation in order to maintain critical concentricity or alignment dimensions.

h. Any critical surface finish or valve body thickness requirements that may affect additive manufacturing of valve bodies. Adjustable set pressure range, required capacity, rated capacity (at accumulation pressure), Maximum allowable blowdown, Blowdown (if adjustable, list adjustable blowdown range as well as actual blowdown setting).

i. Adjustable set pressure range.

j. Required capacity.

k. Rated capacity (at accumulation pressure).

l. Maximum allowable blowdown.

m. Blowdown (if adjustable, list adjustable, list adjustable blowdown range as well as actual blowdown setting)

6.6 Valve orientation. When specified (see 6.2), valves should be packaged with the valve stem vertically upright and packaging marked “This End Up” to avoid damage to sealing surfaces.

6.7 Accumulation test waiver. When specified (see 6.2), the accumulation test (see 4.4.4) may be waived by providing the capacity data, obtained and certified in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII, Rules for Construction of Pressure Vessels.

6.8 Subject term (key word) listing.

Pressure Relief

Safety Relief

Spring Actuated

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy − SH
(Project 4820-2017-013)

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